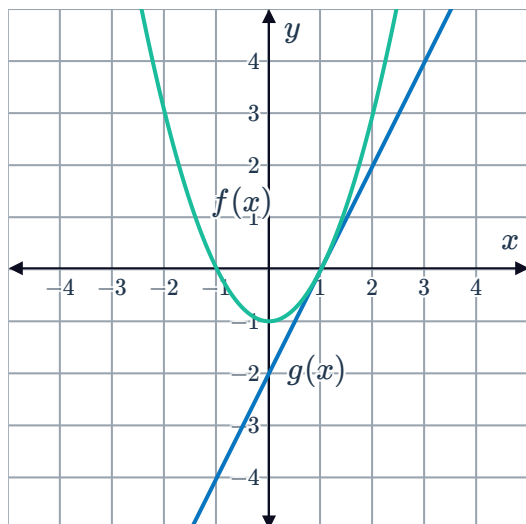


## Equation of a tangent from a graph

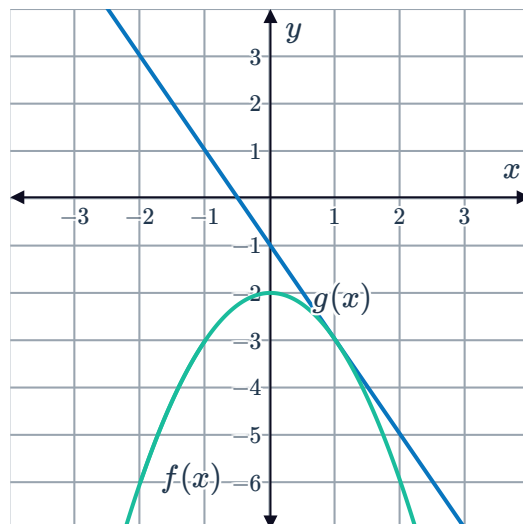


- 1 Each of the following graphs contain a curve,  $f(x)$ , along with one of its tangents,  $g(x)$ .
- State the coordinates of the point at which  $g(x)$  is a tangent to the curve  $f(x)$ .
  - State the gradient of the tangent.
  - Hence determine the equation of the line  $y = g(x)$ .

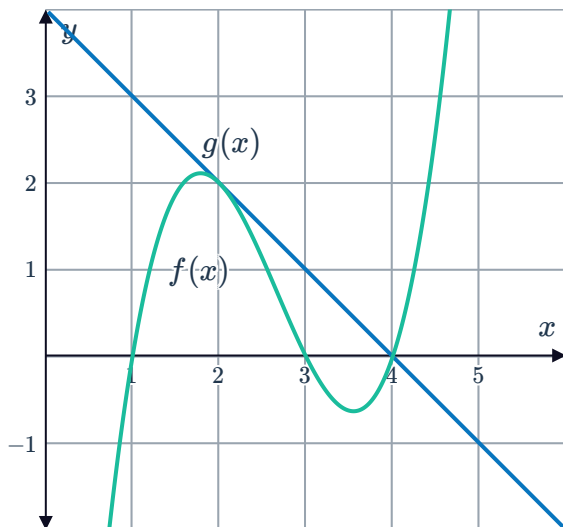
a



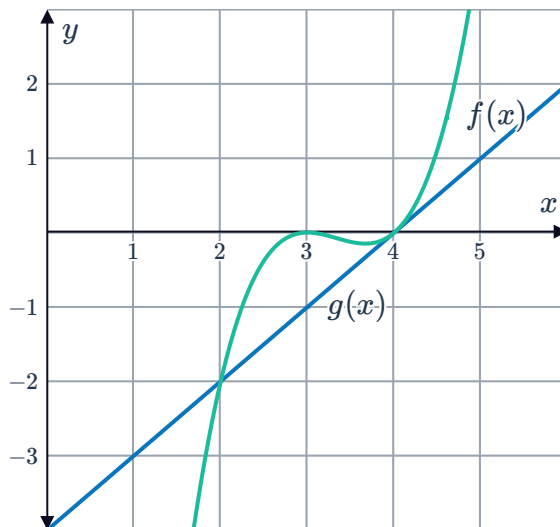
b



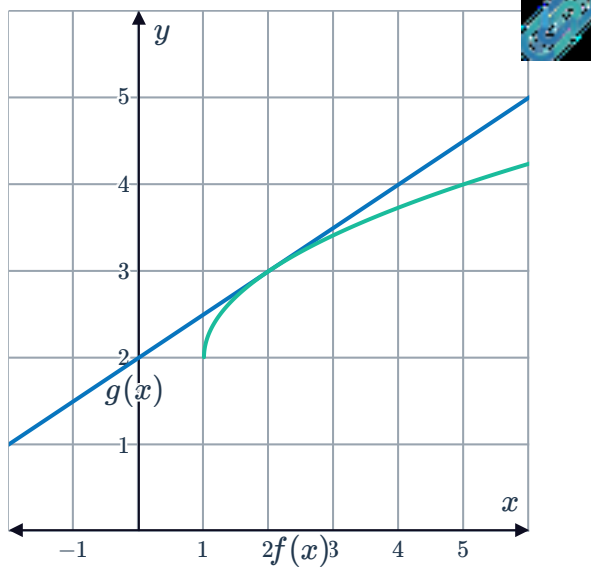
c



d



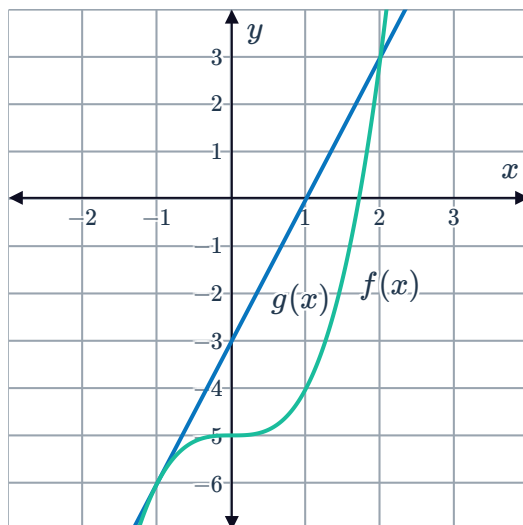
e



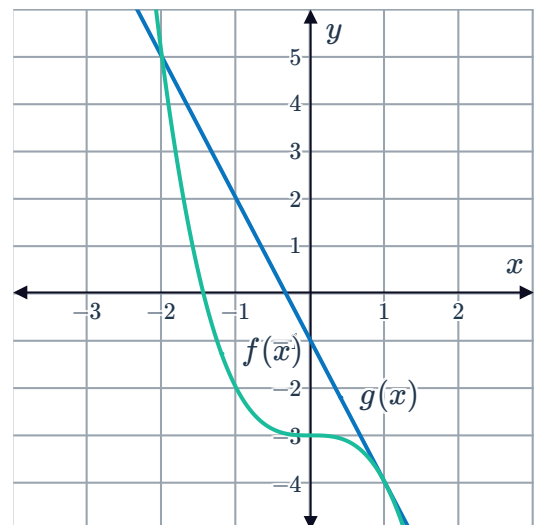
2 Each of the following graphs contain a curve,  $f(x)$ , along with one of its tangents,  $g(x)$ .

- State the coordinates of the point at which  $g(x)$  is a tangent to the curve  $f(x)$ .
- State the gradient of the tangent.
- Hence determine the equation of the line  $y = g(x)$ .
- State the  $x$ -coordinate of the point on the curve at which we could draw a tangent that has the same gradient as  $g(x)$ .

a



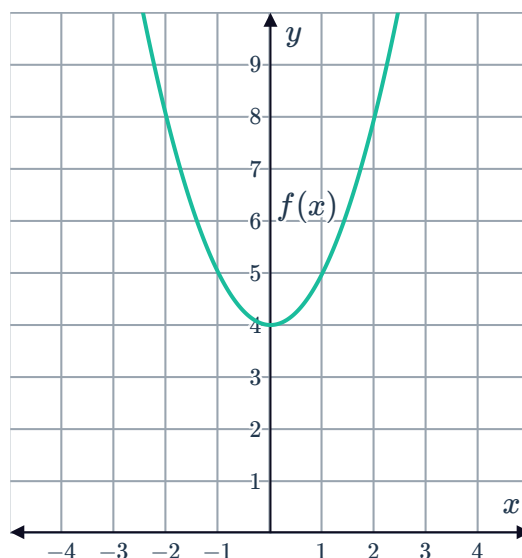
b



3 Consider the graph of the function  $f(x)$ :

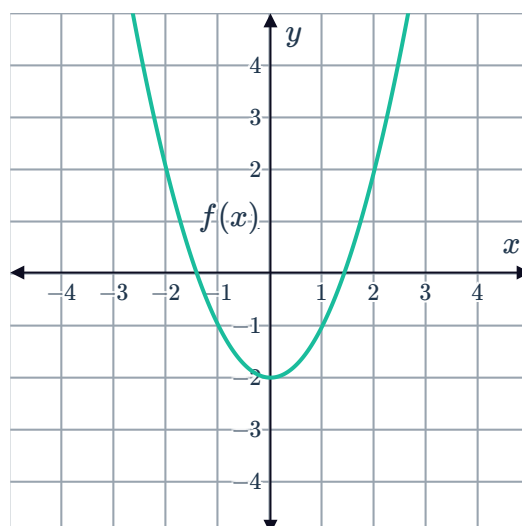


- a Sketch the graph of the function  $g(x) = 2x + 3$  on the same number plane.
- b Is  $g(x)$  a tangent to  $f(x)$ ? Explain your answer.



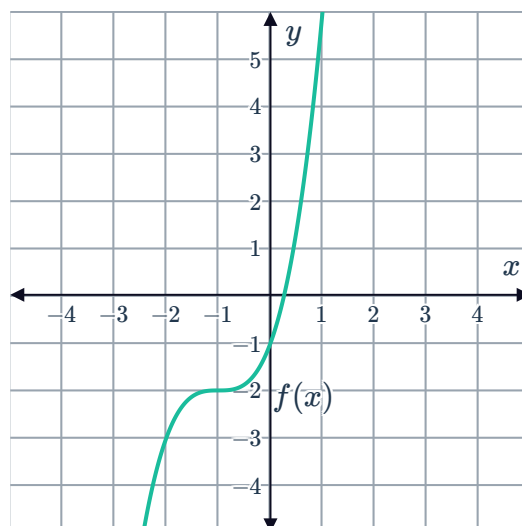
4 Consider the graph of the function  $f(x)$ :

- a Sketch the graph of the function  $g(x) = 2x - 1$  on the same number plane.
- b Is  $g(x)$  a tangent to  $f(x)$ ? Explain your answer.



5 Consider the graph of the function  $f(x)$ :

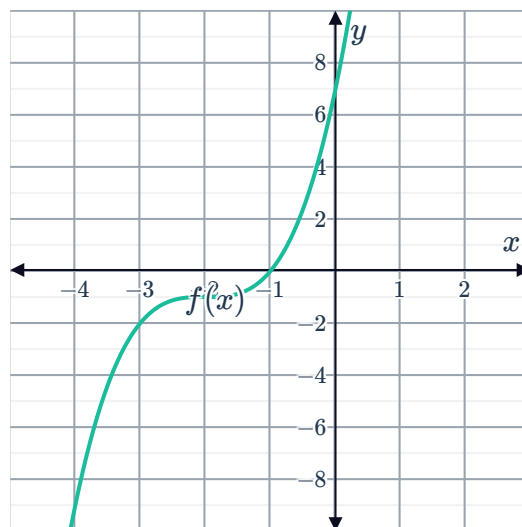
- a Sketch the graph of the function  $g(x) = 3x + 3$  on the same number plane.
- b Is  $g(x)$  a tangent to  $f(x)$ ? Explain your answer.



6 Consider the graph of the function  $f(x)$ :



- a Sketch the graph of the function  $g(x) = 4x + 7$  on the same number plane.
- b Is  $g(x)$  a tangent to  $f(x)$ ? Explain your answer.



7 Consider the curve given by the function  $f(x) = x^2 - 1$ .

- a Find the gradient of the tangent to the curve at the point  $(1, 0)$ .
- b Graph the curve and the tangent at the point  $(1, 0)$  on a number plane.

8 Consider the curve given by the function  $f(x) = x^2 - 4x + 2$ .

- a Find the gradient of the tangent to the curve at the point  $(3, -1)$ .
- b State the coordinates of the vertex of the parabola  $f(x) = x^2 - 4x + 2$ .
- c Graph the curve and the tangent at the point  $(3, -1)$  on a number plane.

## Equation of a tangent at a given point

9 Consider the function  $y = x^2 - 3x + 4$ . State the  $x$ -coordinate of the point on the curve where the tangent makes an angle of  $45^\circ$  with the  $x$ -axis.

10 Consider the tangent to the curve  $f(x) = x^2$  at the point  $(-1, 1)$ .

- a Find the gradient of the function  $f(x) = x^2$  at this point.
- b Find the equation of the tangent to the curve  $f(x) = x^2$  at the point  $(-1, 1)$ .

11 Consider the tangent to the curve  $f(x) = x^3$  at the point  $(-1, -1)$ .

- a Find the gradient of the function  $f(x) = x^3$  at this point.
- b Hence find the equation of the tangent to the curve  $f(x) = x^3$  at the point  $(-1, -1)$ .

- 12 Consider the tangent to the curve  $f(x) = -x^2$  at the point  $(-3, -9)$ .
- Find the gradient of the function  $f(x) = -x^2$  at this point.
  - Hence find the equation of the tangent to the curve  $f(x) = -x^2$  at the point  $(-3, -9)$ .
- 13 Consider the tangent to the curve  $f(x) = x^2$  at the point  $(-2, 4)$ .
- Find the gradient of the function  $f(x) = x^2$  at this point.
  - Find the equation of the tangent to the curve  $f(x) = x^2$  at the point  $(-2, 4)$ .
- 14 Consider the function  $f(x) = 3x^2$ .
- Find the gradient of the function at  $x = 2$ .
  - Find the  $y$ -coordinate of the point on the function at  $x = 2$ .
  - Hence find the equation of the tangent to the curve  $f(x) = 3x^2$  at  $x = 2$ .
- 15 Find the equation of the tangent to the curve  $f(x) = 0.3x^3 - 5x^2 - x + 4$  at  $x = 1$ .

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## Applications

- 16 For each of the following:
- Find the  $x$ -coordinate of point  $M$ .
  - Find the  $y$ -coordinate of point  $M$ .
- At point  $M(x, y)$ , the equation of the tangent to the curve  $y = x^2$  is given by  $y = 4x - 4$ .
  - At point  $M(x, y)$ , the equation of the tangent to the curve  $y = x^3$  is given by  $y = 12x - 16$ .
- 17 Consider the function  $f(x) = \frac{4x^3}{3} + \frac{5x^2}{2} - 3x + 7$ . Find the  $x$ -coordinates of the points on the curve whose tangent is parallel to the line  $y = 3x + 7$ .
- 18  $5x + y + 2 = 0$  is the tangent to the curve  $y = x^2 + bx + c$  at the point  $(9, -47)$ .
- Find the derivative  $\frac{dy}{dx}$  of  $y = x^2 + bx + c$ .
  - Find the gradient of the tangent to the curve at  $x = 9$ .
  - Solve for the value of  $b$ .
  - Solve for the value of  $c$ .

19 Consider the function  $f(x) = x^2 + 5x$ .



- a Find the  $x$ -coordinate of the point at which  $f(x)$  has a gradient of 13.
- b Hence state the coordinates of the point on the curve where the gradient is 13.

20 Consider the function  $f(x) = x^3 - 6x^2$ .

- a Find the  $x$ -coordinates of the points at which  $f(x)$  has a gradient of 495.
- b Hence state the coordinates of the points on the curve where the gradient is 495.

21 Consider the function  $f(x) = x^3 + 3x^2 - 19x + 2$ .

- a Find the  $x$ -coordinates of the points at which  $f(x)$  has a gradient of 5.
- b Hence state the coordinates of the points on the curve where the gradient is 5.

22 Consider the function  $f(x) = x^3 + 6x^2 - 14x - 2$ .

- a Find the  $x$ -coordinates of the points on the curve where the gradient is the same as that of  $g(x) = 22x - 4$ .
- b Hence state the coordinates of the points on the curve where the gradient is the same as that of  $g(x) = 22x - 4$ .

23 Consider the function  $y = 4x^2 - 5x + 2$ .

- a Find  $\frac{dy}{dx}$ .
- b Hence find the value of  $x$  at which the tangent to the parabola is parallel to the  $x$ -axis.

24 The curve  $y = ax^3 + bx^2 + 2x - 17$  has a gradient of 58 at the point  $(2, 31)$ . Find the values of  $a$  and  $b$ .

25 The graph of  $y = ax^3 + bx^2 + cx + d$  intersects the  $x$ -axis at  $(2, 0)$ , where it has a gradient of 36. It also intersects the  $y$ -axis at  $y = -28$ , where the tangent is parallel to the  $x$ -axis.

Find the values of  $a, b, c$  and  $d$ .